Kartik Bali

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PhD candidate applying multimodal LLMs, generative models, and neural simulators to engineering problems: FEM, CAD, and design optimization. First-author papers at **NeurIPS 2026** and **ECCV 2026**. Trained as a mechanical engineer before moving to computational science and differentiable simulation.

Looking for a research internship in Summer or Fall 2027.

Education

- **Ph.D. Candidate, Computer Science**, TU Hamburg & Helmholtz-Zentrum Hereon *Jan 2025 – expected Dec 2027*
  Advisors: Roland Aydin (DFKI) and Robert West (EPFL).
- **M.Sc. Computational Science & Engineering**, Technical University of Munich *Oct 2021 – June 2024*
  Grade 1.5 (German scale, 1.0 is best). Focus on machine learning and simulation. Deutschland Stipendium.
- **B.Eng. Mechanical Engineering**, BITS Pilani, Hyderabad *July 2015 – May 2019*
  Grade 9.63/10. Merit Scholarship (top 2% of students).

Experience

- **Doctoral Researcher**, Helmholtz-Zentrum Hereon, Hamburg *Jan 2025 – Present*
  - **GReFEM** [1]: multimodal LLMs, used zero-shot, mark the stress-critical regions of 3D parts for FEM mesh refinement. With no FEM solve at all, this closes 58% of the gap to a reference-solve oracle.
  - **Look Where You Can** [3]: CAD reconstruction under occlusion. A view-selection policy and a VLM that writes CadQuery code are trained together with RL on execution rewards. +21/+17 effective mIoU over the best baselines, under 1.5% invalid code.
  - Built **MV-GEL** [2]: given a text prompt, it picks views of a CAD mesh and runs a VLM reasoning-segmentation model on them to find the faces and edges the prompt refers to. Up to 1.7× face IoU and 4.5× edge F1 over CLIP-based baselines.
  - **Point2BIM** (ongoing, with DFKI): turning real laser scans into executable BIM programs with a VLM. SFT, then GRPO with execution rewards. Training on 4×H200, inference with vLLM.
  - Co-author on two papers on LLM agents for engineering: automatic discovery of constitutive models [5] and FRAME, an agentic framework for finite element analysis [6].
- **AI Engineer**, Project-S and Beyond Technologies, Munich: ML for radar perception *Sept 2024 – Dec 2024*
- **Master Thesis Intern**, BMW AG, Munich *Sept 2023 – May 2024*
  - Used physics-guided diffusion models and GANs to make the design of 2D/3D flow geometries 2× faster, plugged into an autograd-based 3D topology optimization tool. Thesis grade 1.0.
- **Machine Learning Werkstudent**, TUM Graphics & Visualization (Thuerey group) *May 2022 – Aug 2023*
  - Implemented graph neural networks and convolutional ResNets as neural simulators for chaotic PDEs, wrote CI/CD and tests for PhiFlow, and co-authored the resulting paper [4].
- **Software Engineer**, L&T Construction, Pune: automated data pipelines (Praise Award) *July 2019 – Sept 2021*

Publications

[1] **Bali, K.**, Guru, M. K., Cyron, C. J., and Aydin, R., "GReFEM: Multimodal LLMs as Zero-Shot Semantic Assistants for Physics-Guided 3D Mesh Refinement", *NeurIPS*, 2026 **(accepted, main track)**. arXiv:2607.08798. Code.
[2] **Bali, K.** and Aydin, R., "MV-GEL: Language-Driven Multi-View Geometric Entity Localization on Meshes", *ECCV*, 2026 **(accepted)**. arXiv:2606.31533. Code.
[3] **Bali, K.**, Guru, M., Borjigin, Y., Starostina, A., Cyron, C. J., and Aydin, R., "Look Where You Can: Active View Selection for CAD Reconstruction under Occlusion", under review at *ICLR*, 2027. arXiv:2610.11954.
[4] List, B., Chen, L.-W., **Bali, K.**, Thuerey, N., "Differentiability in Unrolled Training of Neural Physics Simulators on Transient Dynamics", *Computer Methods in Applied Mechanics and Engineering*, 2025.
[5] Tacke, M., …, **Bali, K.**, et al., "Constitutive Scientific Generative Agent (CSGA): Leveraging Large Language Models for Automated Constitutive Model Discovery", *Mach. Learn. Comput. Sci. Eng.*, 2025.
[6] Guru, M. K., Gupta, V., **Bali, K.**, et al., "From Finite Element Analysis to Self-Optimization: FRAME and the Era of Agentic AI in Engineering", under review at *npj Computational Materials*, 2026.
[7] Minder, J., …, **Bali, K.**, …, Aydin, R., West, R., "Synthetic Persona Pretraining: Alignment from Token Zero", preprint, 2026. arXiv:2608.13482.

Skills

- **Domain:** scientific ML, FEM, differentiable simulation, PDEs, topology optimization, physics-guided generative design, CAD
- **ML:** multimodal LLMs, vision-language models, LLM agents, diffusion models, GANs, VAEs, GNNs, neural simulators, RL post-training (GRPO, RLHF), LoRA fine-tuning
- **Computer vision:** 3D reconstruction, multi-view geometry, meshes, point clouds, SDFs, referring/reasoning segmentation
- **Training and inference:** PyTorch (FSDP, DDP), HuggingFace (Transformers, PEFT), vLLM, FlashAttention-2, Docker, Slurm, W&B
- **Tools:** Python, C++, CUDA/Triton, JAX, CadQuery, PyVista, BlenderProc, OpenCV, Git, Linux
- **Languages:** English (C1), German (B1)